

Technical Note TN-2026-09

From Simplicial Minimality to Spectral Dimension: Why $4 \times 2 = 8$ Does Not Yield $d_s = \ln 8$

Miguel Ángel Franco León

Independent Researcher

ORCID: 0009-0003-9492-385X

<https://github.com/MiguelAngelFrancoLeon/mfsu-tetraedro>

12 September 2026

Document type. Technical note (working).

Framework. Tetrahedral Emergent Gravity (TEG) is the programme under examination, not a validated theory.

Primary claim of this note. $d_s = \ln 8$ is a hypothesis. It is not derived from tetrahedral minimality plus intertwiner multiplicity.

Licence. CC BY 4.0.

Keywords: spectral dimension; tetrahedron; SU(2) intertwiners; coarse-graining; emergent gravity; no-go.

Abstract

A non-degenerate simplex in three dimensions has four vertices. Separately, the SU(2)-invariant intertwiner space of a four-valent $j = 1/2$ node is two-dimensional. Their product is eight. It is tempting to set $b_{\text{eff}} = 8$ and $d_s = \ln 8$.

That identification is not a theorem. This note separates three quantities—geometric coordination z , internal multiplicity M , and dynamical branching b_{dyn} —and shows that finite internal multiplicity together with local finite-range dynamics does not change the leading spectral exponent. On a three-dimensional Hausdorff geometry one still has $d_w = 2$ and $d_s = 3$.

If one *assumes* $d_H = 3$ and *requires* $d_s = \ln 8$, the walk dimension must be $d_w = 6/\ln 8 \simeq 2.885$. That is an anomalous-transport target, not a derivation. An alternative programme (reduced Hausdorff dimension $d_H \simeq \ln 8$ with ordinary diffusion) is a different hypothesis and is not treated as proved here.

The missing object is an explicit tetrahedral coarse-graining that couples the intertwiner sector to spatial transport. Until that object exists,

$$4 \times 2 = 8 \not\Rightarrow d_s = \ln 8.$$

Contents

1 Purpose	2
2 What is established	2
2.1 Simplicial minimality	2
2.2 Intertwiner multiplicity	3
3 The tempting step, and why it fails	3
4 Control: local tetrahedral walk	4
5 No-go: finite internal space, local finite range	4
6 Two programmes, both open	4
7 Diagnostic numbers, not TEG constants	5
8 What has been ruled out	5
9 The open problem (the missing bridge)	5
10 Falsifiability	6
11 Status of claims	6
12 Conclusion	6
A Core relations	7

1 Purpose

The purpose of this note is modest: to list what follows from two standard facts, and what does not.

In earlier TEG documents the author used the chain

$$z = 4 \longrightarrow b_{\text{eff}} = 2z = 8 \longrightarrow d_s = \ln 8 \quad (1)$$

and identified $d_s = D_{\text{eff}} = d_f = \ln 8$. That last identification is an ansatz. The present note replaces it with a testable question and records a no-go for the simplest dynamical reading.

Nothing in this text withdraws the public archive (Zenodo / GitHub). It restricts the logical status of one step in that archive.

2 What is established

2.1 Simplicial minimality

A non-degenerate d -simplex has $d + 1$ vertices. For $d = 3$,

$$N_{\text{min}} = 4. \quad (2)$$

The unique regular simplex is the tetrahedron.

This is combinatorial geometry. It is not yet a law for the vacuum, and it does not by itself fix the degree of a physical graph.

Object	Count
Vertices of one 3-simplex	4
Edges incident on one vertex <i>of that simplex</i>	3
Neighbours of a centre whose neighbours form a tetrahedron	4

The third row is a 4-valent *network* picture (centre + tetrahedron of neighbours). It is a further modelling choice. We write

$$z_{\min} = 4 \quad (3)$$

only as a **minimality condition**, not as a derived microscopic Hamiltonian.

2.2 Intertwiner multiplicity

For four spin-1/2 representations,

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1, \quad (4)$$

and a total singlet can be built through two independent channels $(0 \otimes 0) \rightarrow 0$ and $(1 \otimes 1) \rightarrow 0$. Hence

$$\dim \text{Inv}[(\frac{1}{2})^{\otimes 4}] = 2. \quad (5)$$

This is representation theory. It does not depend on a gravitational reading.

A convenient recoupling matrix (phase conventions aside) is

$$F = \begin{pmatrix} -1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{pmatrix}, \quad (6)$$

which is real orthogonal,

$$F^T F = I, \quad \det F = -1, \quad \text{spec}(F) = \{-1, +1\}. \quad (7)$$

Recoupling itself supplies no attenuation.

Remark 1 (Normalisation). *The tensors $\varepsilon_{AB}\varepsilon_{CD}$ and $\varepsilon_{AC}\varepsilon_{BD}$ are linearly independent but not, with identical prefactors, orthonormal. An orthonormal pair is obtained by Gram–Schmidt. The invariant statement is only $\dim \mathcal{H}_{\text{int}} = 2$.*

3 The tempting step, and why it fails

Write $M = \dim \mathcal{H}_{\text{int}} = 2$. Then

$$zM = 8. \quad (8)$$

This counts combinations of geometric directions and internal states. It does not count independent spatial propagation channels. One must distinguish

$$z = \text{geometric coordination}, \quad M = \text{internal multiplicity}, \quad b_{\text{dyn}} = \text{dynamical branching}. \quad (9)$$

The TEG identification would require the extra equality

$$b_{\text{dyn}} = zM = 8, \quad (10)$$

which state counting does not give.

4 Control: local tetrahedral walk

Take the four vectors

$$v_1 = (1, 1, 1), \quad v_2 = (1, -1, -1), \quad v_3 = (-1, 1, -1), \quad v_4 = (-1, -1, 1) \quad (11)$$

and the unit directions $n_i = v_i/\sqrt{3}$. Then

$$\sum_{i=1}^4 n_i^a n_i^b = \frac{4}{3} \delta^{ab}. \quad (12)$$

(The un-normalised v_i give $4\delta^{ab}$, not $4/3$.)

The local Fourier symbol of the graph Laplacian has the small-momentum expansion

$$\lambda(k) = -c k^2 + O(k^4), \quad c > 0. \quad (13)$$

Hence the ordinary local walk has $d_w = 2$. If the large-scale Hausdorff dimension remains $d_H = 3$, the standard relation

$$d_s = \frac{2d_H}{d_w} \quad (14)$$

gives $d_s = 3$. This is the baseline.

5 No-go: finite internal space, local finite range

Enlarge the state space to $\mathcal{H}_{\text{geom}} \otimes \mathbb{C}^M$ with $M < \infty$. Let

$$(T\psi)_\alpha(x) = \sum_{r \in \mathcal{S}} \sum_{\beta} T_{\alpha\beta}(r) \psi_\beta(x+r) \quad (15)$$

be translation invariant, with $\mathcal{S}$ finite. Then $\hat{T}(k)$ is analytic at $k = 0$. If the linear drift vanishes, the critical eigenvalue is generically

$$\lambda_0(k) = 1 - k_a D^{ab} k_b + O(k^4). \quad (16)$$

Isotropy gives $1 - \lambda_0(k) = D|k|^2 + O(k^4)$, so again $d_w = 2$.

Proposition 1 (Structural no-go). *Finite internal multiplicity together with local finite-range dynamics does not imply $d_s \neq d_H$. In particular, two intertwiners do not produce $d_s = \ln 8$.*

The same conclusion holds if the internal transitions are given by $|F_{ij}|^2$, or by any fixed stochastic matrix on $\mathbb{C}^2$. Unitarity of F makes the point sharper: recoupling is a change of internal basis, not a scale-dependent filter.

Assigning finite waiting times to the two internal states, or a finite persistence, can generate *crossovers*. It does not generically produce an anomalous *asymptotic* d_w .

6 Two programmes, both open

Earlier TEG texts identified $d_s = D_{\text{eff}} = d_f = \ln 8$. That collapse mixes two different hypotheses.

Programme A (this note). Keep $d_H = 3$. Then

$$d_s = \ln 8 \iff d_w = \frac{6}{\ln 8} \simeq 2.88539008. \quad (17)$$

Ordinary diffusion is not enough. One needs anomalous transport.

Programme B. Reduce the Hausdorff dimension itself, $d_H \simeq \ln 8$, and keep $d_w = 2$. Then $d_s \simeq \ln 8$ would follow from the same relation, but by a different mechanism (fractal support, not anomalous walk). Programme B is not established by $zM = 8$ either, and is not proved here.

These are not two names for one theorem. A future construction must say which programme it is testing.

7 Diagnostic numbers, not TEG constants

Under Programme A one may parametrize $D(L) \sim L^{-\theta}$ and use $d_w = 2 + \theta$. The target $d_w = 6/\ln 8$ would require

$$\theta = \frac{6}{\ln 8} - 2 \simeq 0.88539008. \quad (18)$$

On dyadic scales $L_n = 2^n$, a phenomenological survival factor $Q(L_n) \sim q^n$ would then have

$$q_* = 2^{-\theta} \simeq 0.54134113. \quad (19)$$

These numbers are **inverse diagnostics**. They are what Programme A would have to *reproduce*. They are not derived from $SU(2)$, and they must not be inserted into a coarse-graining in order to “confirm” $\ln 8$.

No matrix element of F equals q_* . The values $1/4$ and $3/4$ are the recoupling probabilities; they are not a spectral-dimension mechanism.

8 What has been ruled out

1. $z = 4$ alone: local tetrahedral walk, $d_s = 3$ if $d_H = 3$.
2. $z = 4$ plus $M = 2$, still local and finite-range: same leading exponent (Proposition 1).
3. Unitary recoupling: no attenuation factor.
4. Finite internal waiting times: crossovers, not a derived asymptotic $d_s = \ln 8$.
5. Counting $zM = 8$ as if it were b_{dyn} .

Negative controls are the content of this stage.

9 The open problem (the missing bridge)

Define an explicit hierarchy T_n on scales $L_{n+1} = 2L_n$, each copy carrying $\mathcal{H}_{\text{int}} \simeq \mathbb{C}^2$, and a coarse-graining

$$C_n : \mathcal{H}_n \rightarrow \mathcal{H}_{n+1} \quad (20)$$

built from tetrahedral incidence—not from a target number. Measure

$$\sigma_i(C_n), \quad q_{\text{eff}}(L_n), \quad d_w(L), \quad d_s(L). \quad (21)$$

Report whatever emerges.

The question is not whether q_{eff} approaches 0.541. The question is whether *any* independently specified tetrahedral rule produces anomalous transport at all.

Even a positive answer for d_s would still leave gravity open: a spectral dimension is not an effective Einstein equation.

10 Falsifiability

The present form of the $\ln 8$ hypothesis fails if every physically motivated C_n yields

$$d_w \rightarrow 2, \quad d_s \rightarrow d_H. \quad (22)$$

A value near $\ln 8$ obtained only by imposing q_* or θ is not a confirmation.

A genuine confirmation would have the shape

$$\text{microscopic tetrahedral rules, fixed first} \longrightarrow d_w \text{ or } d_H \text{ measured} \longrightarrow d_s. \quad (23)$$

11 Status of claims

Item	Status
$N_{\min} = 4$ for a 3-simplex	Theorem
$\dim \mathcal{H}_{\text{int}} = 2$ for four $j = 1/2$	Theorem
$F^T F = I$	Theorem
Local tetrahedral walk: $d_w = 2$	Theorem (control)
Finite M + local finite range $\not\Rightarrow d_s = \ln 8$	Theorem (no-go)
$zM = 8$ as state count	Definition
$b_{\text{dyn}} = 8$	Not shown
$d_s = \ln 8$	Hypothesis
Programme A ($d_H = 3$, $d_w \simeq 2.885$)	Open
Programme B ($d_H \simeq \ln 8$, $d_w = 2$)	Open, distinct
Effective gravitational dynamics from either programme	Open

12 Conclusion

Two facts stand:

$$3\text{D} + \text{discrete} + \text{simplicial} + \text{minimal} \Rightarrow \text{tetrahedron}, \quad (24)$$

$$\left(\frac{1}{2}\right)^{\otimes 4} \text{ singlets} \Rightarrow \dim \mathcal{H}_{\text{int}} = 2. \quad (25)$$

Their product is eight. Eight is not a spectral dimension:

$$4 \times 2 = 8 \not\Rightarrow d_s = \ln 8. \quad (26)$$

The missing arrow is dynamical: a coarse-graining that turns internal multiplicity into anomalous transport (Programme A), or a derivation of reduced Hausdorff dimension (Programme B). Until one of those arrows is built and tested, $d_s = \ln 8$ remains a hypothesis inside an incomplete framework.

Both a positive and a negative outcome of the next stage are useful. The value of the present note is that the hypothesis is now a precise question rather than an identification.

A Core relations

$$N_{\min} = 4, \tag{27}$$

$$\dim \mathcal{H}_{\text{int}} = 2, \tag{28}$$

$$zM = 8 \quad (\text{count, not } b_{\text{dyn}}), \tag{29}$$

$$d_s = 2d_H/d_w, \tag{30}$$

$$d_H = 3 \text{ and } d_s = \ln 8 \implies d_w = 6/\ln 8 \simeq 2.88539, \tag{31}$$

$$d_w = 2 + \theta \implies \theta \simeq 0.88539, \tag{32}$$

$$q_* = 2^{-\theta} \simeq 0.54134 \quad (\text{diagnostic only}). \tag{33}$$